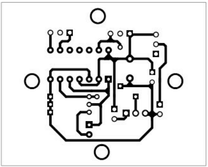

Fig. 3: Actual-size PCB layout of the cistern overflow alert system

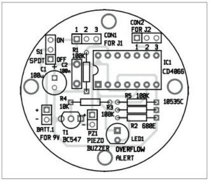

Fig. 4: Component layout of the PCB

(LED1) is driven by IC1A, and the aural indicator (PZ1) is switched on by IC1B with the help of BC547 transistor (T1). IC1C (connected to J2) works as a volt-free normally-open (N/O) bilateral switch for optional remote overflow indication. Maximum current through this switch should be less than 20mA.

As shown in the circuit, three (of the four) electronic switches in CD4066 control device operation. In the case of an overflow, IC1A is closed due to the high level at its control input (pin 13); LED1 lights up to indicate its status. At the same time, IC1B is closed too, causing PZ1 to be active.

In idle state, voltage at the control inputs of all switches in IC1 drops to zero, causing the switches to open. Overflow detection threshold may be adjusted by using appropriate value of resistor R1. Switch S1, when opened, prevents the battery from being discharged when the device is not in use for some reason.

Further, output from the active piezo buzzer should be loud enough to alert you. If desired, a more powerful piezo sounder can be used instead.

Construction and testing

An actual-size, single-side PCB layout of the cistern overflow alert system is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the PCB.

Use a 9V PP3 battery or suitable DC power supply. Connect the cistern sensors to the PCB at CON1 using two wires (J1). Connect two copper wires at CON2 for extension to optional remote indication (detail not shown here). Place the assembled unit at a suitable location for monitoring water overflow condition. EFY

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